Spectral–Spatiotemporal Prompting for Multispectral UAV Tracking

Xiaoyue Yan, Tingfa Xu, Haolin Qin, Tianhao Li, and Jianan Li

***Abstract*—UAV tracking often faces critical challenges such as target appearance variations and poor lighting conditions. Compared to RGB images, multispectral images (MSIs) possess rich spatial–spectral information, enabling trackers to better handle scenarios involving illumination variations and similar colors. Existing multispectral (MS) tracking methods typically rely on full fine-tuning of RGB-based models and inadequately exploit temporal information, leading to suboptimal performance due to limited training data. To address these issues, we propose a unified spectral–spatiotemporal prompting (SSTP) for MS UAV tracking, named SSTP. We design a dual-granularity spatial–spectral fusion (DGS^{2}F) module to extract and fuse coarse-grained spatial features and fine-grained spectral features, enabling more comprehensive utilization of detailed spectral information. Furthermore, we introduce hierarchical temporal modeling, where short-term memory prompt (SMP) enhances the backbone network for joint spectral–spatiotemporal feature extraction, while Mamba-based long-term memory prompt (MLMP) facilitates long-range motion sequence modeling. Extensive experiments on the MS tracking benchmarks demonstrate that our method achieves state-of-the-art performance while maintaining real-time efficiency, with separate training and evaluation conducted on each benchmark, achieving an AUC of 69.2% on the MUST dataset and leading performance on both the HOT2020 and HOT2024 datasets. The code will be available at https://github.com/YXYbit/SSTP**

***Index Terms*—Adaptive learning, hierarchical temporal modeling, multispectral (MS) object tracking, spectral–spatiotemporal prompting (SSTP), UAV tracking.**

I. Introduction

ROBUST object tracking is critical for intelligent UAV systems, yet practical deployment faces significant challenges, including small targets, poor illumination, and similar color [1], [2], [3], [4]. These factors degrade spatial feature extraction, substantially limiting the performance of RGB trackers. Multispectral images (MSIs) offer complementary spectral signatures that capture the material and reflective properties of targets, providing enhanced stability and robustness in dynamic environments. The MUST dataset [5] is the first large-scale, challenging benchmark for multispectral (MS) UAV tracking, offering essential data support for real-world scenarios. With the advancement and miniaturization of MS cameras, MSI-based vision tasks have gained increasing attention.

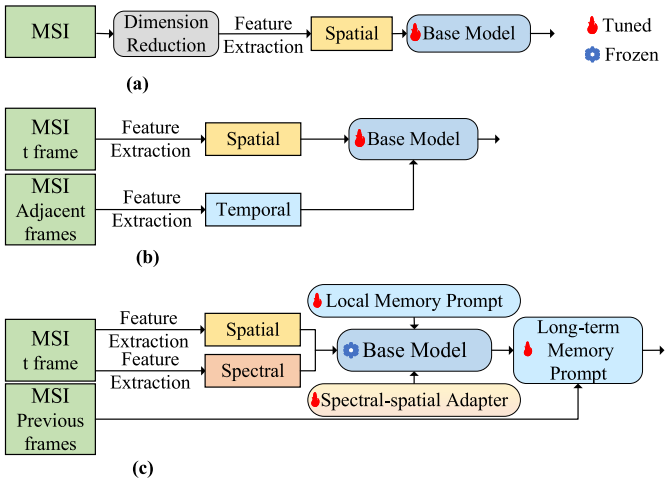


Fig. 1. Illustration of different MS tracking paradigms. SSTP achieves more robust tracking by fully utilizing spatial–spectral information and incorporating richer temporal cues. (a) Full fine-tuning method with dimension reduction. (b) Spatiotemporal fusion with adjacent frames. (c) Joint SSTP method.

Early approaches relied on hand-crafted features extracted from MSIs to capture material characteristics, which were then embedded into correlation filters for tracking. With the advent of deep learning, more MS trackers have adopted deep learning-based architectures capable of extracting high-level semantic information. Nevertheless, MS tracking lacks large-scale datasets comparable to those available for general RGB tracking, which limits the training of deep learning networks that rely on massive amounts of data. Consequently, these trackers often exhibit degraded performance in complex scenarios and insufficient efficiency to meet the dual requirements of accuracy and real-time performance in UAV tracking.

MSIs contain multiple spectral channels, making pretrained RGB trackers unsuitable for direct application. As shown in Fig. 1(a), band dimensionality reduction methods typically convert an MSI into one or more three-channel images, which are then fed into an RGB tracker for fully fine-tuning (FFT). Some MS trackers obtain a false-color image

Received 28 February 2026; revised 10 June 2026 and 23 July 2026; accepted 20 August 2026. Date of publication 25 August 2026; date of current version 3 September 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62101032, in part by Chongqing Excellent Young Scientists Fund under Grant CSTB2025NSCQ-JQX0017, in part by the High-Quality Development Special Project of the Ministry of Industry and Information Technology under Grant TC240HAJ9-35, and in part by Beijing Nova Program under Grant 20240484543. *(Corresponding authors: Tingfa Xu; Jianan Li.)*

Xiaoyue Yan, Haolin Qin, and Tianhao Li are with Beijing Institute of Technology, Beijing 100081, China (e-mail: 3120240591@bit.edu.cn; 3120225333@bit.edu.cn; 3220225103@bit.edu.cn).

Tingfa Xu and Jianan Li are with Beijing Institute of Technology, Beijing 100081, China, and also with Chongqing Innovation Center, Beijing Institute of Technology, Chongqing 401135, China (e-mail: ciom_xtf1@bit.edu.cn; lijianan@bit.edu.cn).

Digital Object Identifier 10.1109/TGRS.2026.3727459

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:20:28 UTC from IEEE Xplore. Restrictions apply.

by bands extraction [6], [7], [8], while others generate multiple false-color images via band grouping [9] or modality synthesis [10], followed by feature-level [11] or decision-level fusion [12], [13]. However, such methods remain inherently suboptimal. First, the band dimensionality reduction process inevitably disrupts the continuity and completeness of spectral responses. This limits the tracker's ability to fully exploit the rich spatial–spectral information in MSIs. Second, full fine-tuning requires updating a large number of parameters, making the tracking model prone to overfitting under limited MS training data.

In addition, most existing methods focus on extracting spatial–spectral features, while the exploration of joint spectral–spatial–temporal modeling remains limited. SP-HST [14] and Trans-DAT [15] use only the initial template and current search region, neglecting temporal memory. SPIRIT [11] and SSTFT [16] introduce dynamic templates to adapt to target appearance variations. However, these methods typically rely on manually designed update mechanisms and hyperparameters (e.g., thresholds or intervals). OCSCNet [17] and HDSP [18] both incorporate information from historical frames to assist in modeling the current frame, as illustrated in Fig. 1(b), but they only exploit adjacent frames, without fully capturing or modeling the target's temporal motion information.

Recently, parameter-efficient fine-tuning (PEFT) provides a promising alternative to mitigate the data scarcity issue by adapting pretrained foundation models with only a small number of trainable parameters. Its effectiveness has been demonstrated in RGB [19] and multimodal tracking [20], [21], [22], [23], where prompts or adapters are used to introduce historical cues or auxiliary modality information. Inspired by these successes, recent MS trackers have adopted PEFT to better transfer RGB tracking knowledge to MS tracking [14], [24], [25], [26]. Nevertheless, most of them primarily focus on spectral or spatial adaptation, while temporal information remains insufficiently exploited.

Current solutions for MS UAV tracking still have significant limitations.

1) *Inadequate Spectral Bands Processing:* Band dimensionality reduction disrupts spectral continuity and completeness, while band grouping introduces information redundancy and higher computational complexity.
2) *Limited MS Training Data:* Limited training data make full fine-tuning prone to overfitting and suboptimal performance.
3) *Insufficient Temporal Modeling:* Insufficient temporal information exploitation limits robustness to handle complex scenarios such as rapid object changes and occlusion.

Hence, we propose SSTP, as shown in Fig. 1(c), an MS UAV tracker based on spectral-spatiotemporal joint prompting. Different from existing MS trackers, SSTP reformulates MS tracking as a prompt-guided adaptation problem. In particular, spectral details and temporal cues are encoded as complementary prompts to adapt a pretrained RGB tracker to MS scenarios with limited training data. To fully exploit spectral details, we introduce a dual-granularity spatial–spectral fusion (DGS^{2}F) module at the feature extraction stage. DGS^{2}F preserves RGB-pretrained parameters to extract coarse-grained spatial features from false-color images while exploiting fine-grained spectral information from the original MSI as prompts to guide spatial–spectral feature fusion, thereby enabling more comprehensive utilization of spectral information rather than compressing MSIs into three-channel inputs.

To fully exploit temporal information, we design a hierarchical memory prompting mechanism. In particular, the short-term memory prompt (SMP) introduces local temporal cues into the backbone in a lightweight manner, enabling spectral–spatiotemporal joint feature extraction for robust representation. Subsequently, a Mamba-based long-term memory prompt (MLMP) is employed to model long-range temporal dependencies. By token propagation, this module extends the memory horizon back to the initial frame, providing more stable and informative features to guide subsequent tracking.

We conducted a comprehensive evaluation of the proposed SSTP and various prompt-learning-based trackers on the MS UAV tracking dataset. Experimental results demonstrate that SSTP achieves state-of-the-art performance. Furthermore, we extend our experiments to other MS general object tracking datasets, which further validate the superiority and efficiency of SSTP. In summary, the main contributions of this work are as follows.

1) We formulate MS UAV tracking as a unified spectral–spatiotemporal prompting (SSTP) problem and propose SSTP, a prompt-based adaptation framework that jointly exploits spectral, spatial, and temporal cues. SSTP achieves outstanding performance on the MS UAV tracking dataset.
2) We design a DGS^{2}F module that fully exploits spatial–spectral information by jointly modeling coarse-grained spatial features with fine-grained spectral details.
3) We introduce a hierarchical memory prompting mechanism for temporal modeling. The SMP enhances spectral–spatiotemporal joint feature extraction, while the MLMP models global target motion evolution, improving robustness in challenging UAV scenarios.

II. Related Work

A. MS UAV Tracking

In RGB object tracking, Siamese-based tracking frameworks have dominated over the past few years, formulating the tracking task as a template similarity matching problem [27]. Building on this foundation, SiamRPN [28] decouples the matching problem into two subtasks, foreground and background classification and target box regression, which enhances the performance of the tracker. With the development of transformer architectures in visual tasks, one-stream architectures [29] have gradually replaced Siamese approaches due to their efficiency in feature modeling and association. In UAV tracking, most existing methods are built on traditional correlation filters [30] or CNN-based Siamese architectures. Recently, ViT-based trackers have shown the potential of transformer architectures in UAV scenarios. FCADtrack [31] further enhanced ViT by introducing spatial context awareness

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:20:28 UTC from IEEE Xplore. Restrictions apply.

and UAV-specific adaptation. Nevertheless, ViT-based UAV trackers remain relatively limited and are mostly based on RGB data, which exhibits performance limitations under challenging scenarios [32].

HOT [33] is the first MS object tracking dataset, primarily focusing on generic targets in real-world scenes, and a series of MS trackers have been developed based on it. Early trackers primarily adopted traditional methods, such as correlation filters, which embedded spectral information into the filters through hand-crafted feature design [33], [34], [35]. With the rise of deep learning, most MS trackers perform band dimensionality reduction to enable MSI to undergo full fine-tuning based on RGB trackers. BAHT [6] and SiamTU [36] employ background-aware and filtering techniques, respectively, to select the most target-salient spectral bands, while BS-SiamRPN [7] and UBSTrack [8] utilize entropy-based computation and attention mechanisms to choose the bands containing the most informative content. In addition, band-regrouping-based methods decompose MSI into multiple three-channel images and integrate multiple features or tracking results through various fusion strategies [10], [11], [12], [13]. Adapting existing visual models to new domains has also been investigated under domain shifts, where domain adaptation methods are developed to alleviate distribution discrepancies across different visual domains [37], [38]. As PEFT has demonstrated its superiority in downstream tasks with limited data, several studies have also adopted this framework to adapt to various types of MS data with minimal training cost [14], [24], [25], [26].

Overall, existing MS trackers can be roughly categorized into three paradigms: traditional methods, FFT methods based on RGB trackers with band dimensionality reduction or regrouping, and PEFT-based methods. Despite progress, existing methods still suffer from incomplete utilization of spatial–spectral characteristics, redundant architectures, and underexplored temporal cues. These limitations are particularly critical for MS UAV tracking, where handling appearance variations and fast motion requires efficient joint modeling of spectral, spatial, and temporal cues.

B. Parameter Efficient Fine-Tuning in Tracking

PEFT has emerged as a novel training paradigm that significantly improves the performance of numerous downstream natural language processing tasks, owing to its lightweight computational and storage requirements [39]. Representative PEFT techniques include adapter tuning, prompt learning, and low-rank adaptation (LoRA), all of which adapt pretrained models with a small number of additional or updated parameters [40], [41], [42].

PEFT has demonstrated significant advantages in data-limited visual tasks [43], particularly in multimodal tracking, such as RGB-T [44], RGB-D, and RGB-E tracking, where prompts or adapters are often used to reduce modality gaps and exploit complementary information from heterogeneous sensors. Recent methods further extend multimodal tracking to RGB-N and RGB-M tracking. OneTracker introduces unified prompt embedding and uses BERT [45] to extract linguistic features, while UETrack [46] further improves unified tracking through token-pooling mixture-of-experts and target-aware adaptive distillation. Nevertheless, these unified trackers primarily focus on generic tracking and do not explicitly exploit spectral information or effectively utilize temporal cues. MS tracking shares a similar motivation, as MSIs can be viewed as a combination of spatial information and complementary spectral cues. Accordingly, recent studies have adopted PEFT to inherit the generalization capability of RGB trackers, thereby addressing the data heterogeneity and scarcity issues in MS tracking. DaSSP-Net [25] introduces a band-invariant spatial–spectral prompt module that guides the foundation model in learning a universal representation across all domains. SSTtrack [10] and SpectralTrack [47] generate multiple modalities from MSIs and construct spectral–spatial adapters to enable dynamic transfer and interaction across modalities. However, such methods inevitably introduce additional computational overhead and structural redundancy. Therefore, effectively integrating spectral, spatial, and temporal cues within a lightweight adaptation framework remains a key challenge for MS UAV tracking.

C. Mamba in Visual Tasks

The state space model (SSM) was originally proposed for sequence modeling tasks in natural language processing and has shown great potential in addressing long-range dependence problems [48]. Mamba makes the parameters of SSM a function of the input, effectively improving the model's ability to perform content-based reasoning. It has achieved state-of-the-art performance across various sequence modeling tasks, including speech and audio processing [49]. Mamba model has demonstrated exceptional semantic modeling capabilities in the field of natural language processing, and it has also been widely adopted in vision tasks due to its linear computational complexity. For instance, Vim [50] proposed a bidirectional state space mechanism, and Vmamba [51] introduced a four-way scanning strategy. All of these efforts constructed vision backbones with linear time complexity and achieved strong performance across various visual perception tasks. Moreover, MambaVision [52] combined the ViT architecture with Mamba to further enhance the modeling of long-range spatial dependencies.

Several studies have also applied Mamba to object tracking tasks, leveraging its hidden states to continuously record and transmit abundant contextual information from video streams, thereby enabling more robust target tracking [53], [54]. MambaTrack introduces a Mamba-based motion predictor that captures multiobject motion patterns to achieve precise trajectory prediction [55]. To fully leverage the autoregressive property of Mamba and its strength in long-sequence modeling, we innovatively incorporate it into MS UAV tracking, enabling long-range modeling of spectral target motion sequences.

III. Method

A. Overview

The framework of the SSTP is shown in Fig. 2. It consists of three parts: an adaptive spectral–spatiotemporal backbone network for joint feature extraction, a Mamba-based long-term memory prompting module for comprehensive temporal modeling, and a prediction head. The tracker accepts a pair

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:20:28 UTC from IEEE Xplore. Restrictions apply.

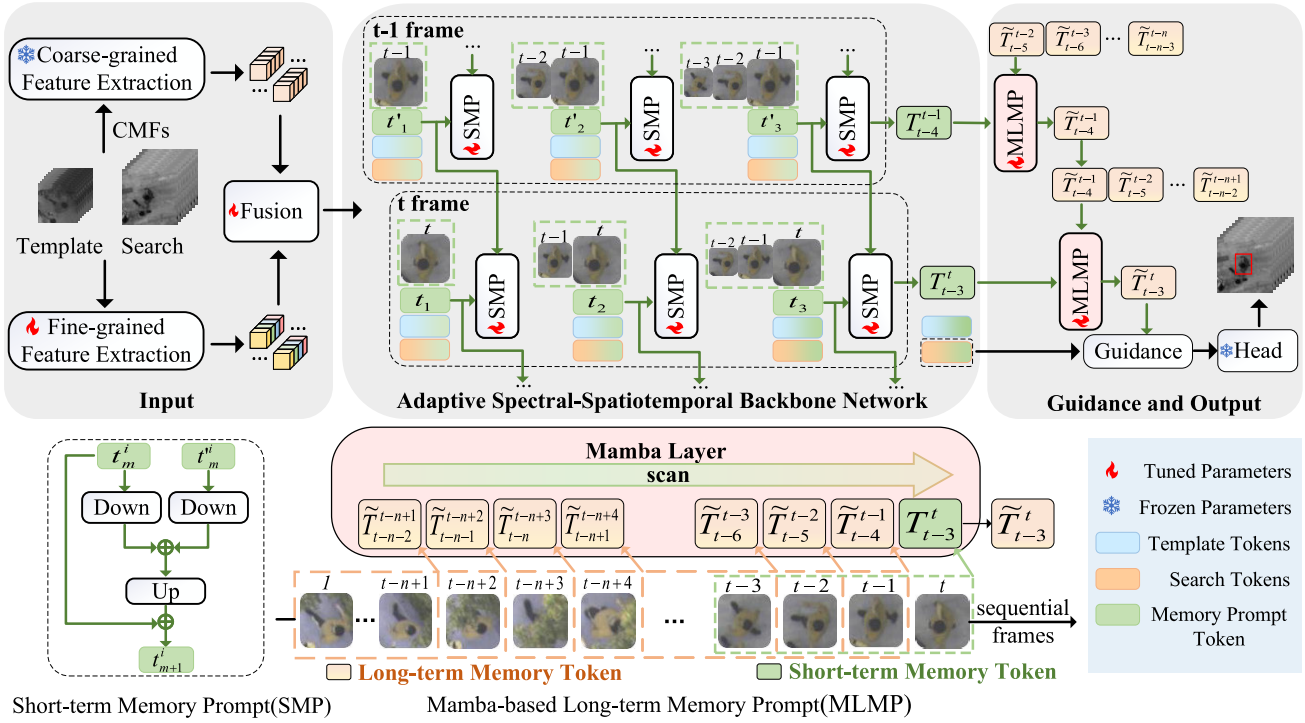


Fig. 2. Overall framework of SSTP. SSTP takes MS template and search images as input, and produces tracking outputs through joint spectral–spatiotemporal feature extraction and long-term memory guidance. The input to the SMP includes local memory tokens from both the current and previous frames.

of MSI as input, comprising a template image $\mathbf{Z} \in \mathbb{R}^{H_z \times W_z \times 8}$ and a search image $\mathbf{X} \in \mathbb{R}^{H_x \times W_x \times 8}$. First, we extract false-color images using CIE color matching functions (CMFs) and process dual-granularity patch embedding to generate spatial–spectral fused features. These tokens are concatenated with a memory prompt token and subsequently fed into the backbone. Subsequently, historical short-term prompt tokens are progressively introduced at different stages of the backbone network, enabling joint spectral–spatiotemporal modeling with gradually increasing memory spans. Finally, the short-term prompt token is processed by the Mamba-based long-term memory prompting module to generate a persistent long-term prompt token, which is propagated across frames. Under the guidance of the long-term prompt token, the search features are processed by the prediction head to yield the final prediction.

B. Adaptive Spectral–Spatiotemporal Joint Feature Extraction

SSTP adopts an adapter tuning strategy to transfer pretrained knowledge to MSI, instead of full fine-tuning. This strategy better leverages the rich knowledge from a pretrained RGB-based model to address the scarcity of training data. Inspired by OSTrack, which demonstrates that a compact single-stream architecture can jointly perform feature extraction and feature association for both the target and search images, we build our baseline upon this framework. We first employ CMFs to generate false-color images from MSI, which are then processed through our dual-granularity feature extraction module to obtain spatial–spectral fused representations $\mathbf{Z}_1^0 \in \mathbb{R}^{N_z \times D}$ and $\mathbf{X}_1^0 \in \mathbb{R}^{N_x \times D}$. Here, $N_z = H_z W_z / 16^2$, $N_x = H_x W_x / 16^2$, $D = 768$, and 0 denotes the backbone

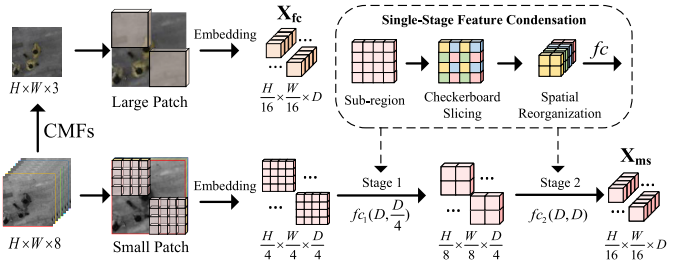


Fig. 3. Processing in DGS^{2}F. The coarse-grained spatial features and fine-grained spectral features are subsequently fused and then fed into the backbone network.

layer index, while 1 indicates the memory span length. To efficiently incorporate temporal information, we introduce a memory prompt token $\mathbf{t}_1^0 \in \mathbb{R}^{1 \times D}$ for each frame. The input to the backbone network can be formally expressed as follows:

$$\mathbf{F}_1^0 = \text{Concat}\left(\mathbf{t}_1^0, \mathbf{Z}_1^0, \mathbf{X}_1^0\right). \tag{1}$$

1) Dual-Granularity Spatial–Spectral Fusion: To fully exploit spatial–spectral information, DGS^{2}F implements dual-granularity feature extraction. For false-color images, we employ pretrained 16×16 patch embeddings to capture coarse-grained spatial features $\mathbf{Z}_{\mathbf{fc}}$ and $\mathbf{X}_{\mathbf{fc}}$, while for MSI, 4×4 patch embeddings extract fine-grained spectral details, generating N_z and N_x subregions (each comprising 4×4 patches), respectively. As shown in Fig. 3, each subregion undergoes two-stage checkerboard slicing and spatial reorganization. Both operations are parameter-free and effectively rearrange features by aggregating them along the channel dimension.

MSIs contain multiple, or even dozens of, spectral bands. Directly increasing the number of input channels in the convolutional layer would significantly raise the parameter burden

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:20:28 UTC from IEEE Xplore. Restrictions apply.

of the first layer. In the design of DGS^{2}F, the embedding layer employs small convolutional kernels with an output dimension of $D/4$, which enables fine-grained feature extraction while simultaneously reducing the number of parameters in the first layer. In the first stage, the fc_1 layer following the first reorganization filters valuable spatial and channel information, while the subsequent fc_2 layer in the second stage performs semantic alignment without altering the feature dimensionality. The subregions are ultimately compressed to 1×1, yielding $\mathbf{Z_{ms}} \in \mathbb{R}^{N_z \times D}$ and $\mathbf{X_{ms}} \in \mathbb{R}^{N_x \times D}$ that maintain dimensional consistency with $\mathbf{Z_{fc}}$ and $\mathbf{X_{fc}}$. Finally, the two sets of features are aligned and fused, as formally expressed by

$$\begin{aligned} \mathbf{Z}_1^0 &= f_3\left(f_1\left(\mathbf{Z_{fc}}\right) + f_2\left(\mathbf{Z_{ms}}\right)\right) \\ \mathbf{X}_1^0 &= f_3\left(f_1\left(\mathbf{X_{fc}}\right) + f_2\left(\mathbf{X_{ms}}\right)\right) \end{aligned} \tag{2}$$

where the projection functions $f(\cdot)$ are 1×1 convolutional layers, $f_1(\cdot)$ and $f_2(\cdot)$ map the coarse-grained $\mathbf{X_{fc}}$ and fine-grained $\mathbf{X_{ms}}$ features to a lower dimension D/α for alignment, and α is set to $768/8$. $f_3(\cdot)$ projects the fused features back to the original high-dimensional representation.

2) Adapter-Based Transformer: MS data vary significantly in their spectral coverage and number of channels across different imaging systems. If full fine-tuning of foundation models were performed for each type of data, it would require learning a new set of parameters equal in size to the original model. The parameters of the foundation tracker are denoted as θ. For a given MS dataset, assuming the optimal parameter weights are θ', the objective of full fine-tuning is to learn the discrepancy between θ' and θ, denoted as $\Delta\theta$, where $\Delta\theta$ has the same parameter dimensionality as θ. Due to the limited scale of each MS dataset compared to the abundance of RGB data, this approach could lead to suboptimal performance or even catastrophic forgetting.

Unlike approaches that only introduce learnable parameters at the input space of transformer layers, we design a more suitable adaptation method tailored for transformer architectures. The transformer architecture fundamentally relies on the attention mechanism as its computational core, where the associated linear layers encode rich feature representations. To leverage this structure, we propose targeted adaptations to the query, key, value, and output components within the attention mechanism. As shown in Fig. 4, we insert trainable adapters with a small number of parameters after the linear projection operation. Taking query as an example, $\mathbf{Q} \in \mathbb{R}^{c \times l}$ is adjusted to $\mathbf{Q}' \in \mathbb{R}^{c \times l}$, the process becomes

$$\mathbf{Q}' = \mathbf{Q} + W_{up} GELU\left(W_{down}\mathbf{Q}\right) \tag{3}$$

where $W_{down} \in \mathbb{R}^{r \times c}$ and $W_{up} \in \mathbb{R}^{c \times r}$ are two mapping matrices, $GELU$ is gelu operation, and rank $r \ll c$. During finetuning, we only train W_{down} and W_{up} to achieve efficient learning of $\Delta\theta$.

3) Short-Term Memory Prompting: Temporal information, which records target evolution across consecutive frames, constitutes a valuable cue for visual tracking. Most existing MS trackers either neglect temporal modeling or handle it in a limited manner. Our framework addresses this by progressively incorporating temporal information during feature encoding through memory prompt tokens.

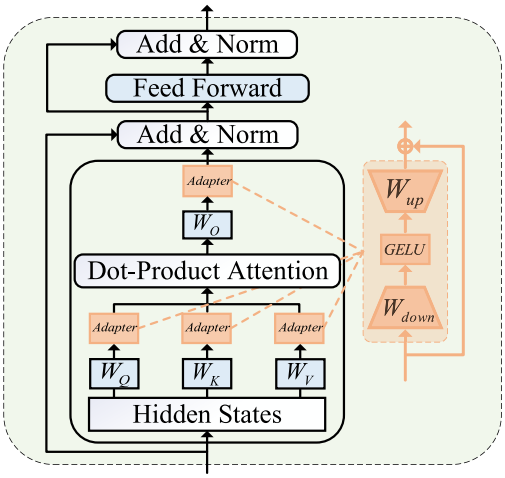


Fig. 4. Adapter-based transformer layers. The standard adapters are inserted into the transformer layers in a residual structure, with only these parameters being trainable.

SMP injects local temporal cues into the backbone by updating the memory prompt token with its counterpart from the previous frame. In particular, at the three candidate elimination layers of the backbone, the current memory token t_m^i interacts with the previous-frame token $t_m'^i$

$$t_{m+1}^i = \mathrm{SMP}\left(t_m^i, t_m'^i\right), \quad i \in \{3, 6, 9\} \tag{4}$$

where i denotes the index of the backbone layer and m denotes the memory span. The specific computations within the SMP module are formally expressed as follows:

$$t_{m+1}^i = t_m^i + g_2\left(g_1\left(t_m^i\right) + g_1\left(t_m'^i\right)\right) \tag{5}$$

where the $g_1(\cdot)$ and $g_2(\cdot)$ are linear layers, $g_1(\cdot)$ reduces the feature dimension to D/β, $g_2(\cdot)$ restores it to D, and β is set to $768/8$. The memory span m is initialized as 1 and progressively increases to 2, 3, and 4 through three temporal interaction stages. This means that for a current frame t, the feature extraction process leverages target information from frames $t-1$, $t-2$, and $t-3$.

Through attention mechanisms, the feature representation incorporates richer temporal information. Moreover, as the network deepens, the memory prompt tokens progressively distill essential information from the current frame. The temporal prompt tokens $\mathbf{T}_{t-3}^t$ are preserved for subsequent processing in the long-term memory module; t represents the current frame, and $t-3$ indicates memory extension spanning back to the $t-3$ frame.

C. Mamba-Based Long-Term Memory Prompting

Building upon the SMP that captures nearby frame interactions, we further propose the MLMP module for long-range temporal modeling to more comprehensively utilize temporal information. Mamba demonstrates exceptional capability in processing sequential data with long-range dependencies, particularly temporal sequences.

The selective SSM serves as a fundamental component of Mamba, which recursively maps the input sequence $x(t) \in \mathbb{R}$ to the output sequence $y(t) \in \mathbb{R}$ through hidden state transitions.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:20:28 UTC from IEEE Xplore. Restrictions apply.

A continuous-time SSM can be formally characterized by its state equation and output equation

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t)$$
$$y(t) = \mathbf{C}h(t) + \mathbf{D}x(t) \tag{6}$$

where $h(t)$ is the hidden state, $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$ are the weighting parameters. For practical implementation of the Mamba architecture within deep learning frameworks, the continuous parameters $\mathbf{A}$ and $\mathbf{B}$ require discretization. The zero-order hold (ZOH) technique, a standard discretization approach, transforms these parameters as defined below

$$\overline{\mathbf{A}} = \exp(\mathbf{\Delta A})$$
$$\overline{\mathbf{B}} = (\mathbf{\Delta A})^{-1}\left(\exp(\mathbf{\Delta A}) - \mathbf{I}\right) \cdot \mathbf{\Delta B} \tag{7}$$

where Δ represents the discretization timescale parameter, the transformed system equations become

$$h_t = \overline{\mathbf{A}}h_{t-1} + \overline{\mathbf{B}}x_t$$
$$y_t = \mathbf{C}h_t + \mathbf{D}x_t \tag{8}$$

where the parameters Δ, $\mathbf{B}$, and $\mathbf{C}$ are input-dependent, enabling dynamic filtering of irrelevant sequence elements through context-aware state transitions.

At frame t, MLMP takes the short-term token T^t_{t-3} produced by the backbone and a set of historical long-term tokens $\widetilde{T}$ as input

$$T_{\text{in}} = \begin{cases} \text{Concat}\left(\widetilde{T}^1_1, \ldots, T^t_{t-3}\right), & t < n \\ \text{Concat}\left(\widetilde{T}^{t-n+1}_{t-n-2}, \ldots, T^t_{t-3}\right), & t \geq n \end{cases} \tag{9}$$

where n is the temporal window length. The token sequence is then processed by an Mamba layer

$$T_{\text{out}} = \text{MLMP}(T_{\text{in}}). \tag{10}$$

The last token of T_{out}, denoted as $\widetilde{T}^t_{t-3}$, is propagated to the next frame and used as the long-term memory prompt for the current frame. Due to the autoregressive property of Mamba and the token propagation mechanism in MLMP, each long-term output prompt token $\widetilde{T}$ encapsulates complete historical target information from the initial frame to the current frame, thus achieving extended temporal sequence modeling.

D. Guidance and Learning Objective

The long-term memory prompt token $\widetilde{\mathbf{T}}^t_{t-3} \in \mathbb{R}^{1\times D}$ of the current frame distills critical information spanning from the initial frame to the present. We employ this token in a prompting manner to guide the short-term spectral–spatiotemporal joint search features $\mathbf{X} \in \mathbb{R}^{N_x \times D}$ output by the backbone, thereby achieving more robust target localization. The detailed operation is formally expressed as follows:

$$\widetilde{X} = X + X \odot \left(\widetilde{\mathbf{T}}^t_{t-3}\right)^{\top} \tag{11}$$

where $\odot$ means the dot product. $\widetilde{X}$ is subsequently processed by the prediction head. In SSTP, the head architecture and loss function maintain consistency with the foundation model, employing a center-based head with dual branches for classification and regression. The loss function is formally defined as follows:

$$\mathcal{L} = \mathcal{L}_{cls} + \lambda_{giou}\mathcal{L}_{giou} + \lambda_{\mathcal{L}_1}\mathcal{L}_1 \tag{12}$$

TABLE I

COMPARISON WITH STATE-OF-THE-ART TRACKERS ON THE MUST DATASET. UNLESS OTHERWISE SPECIFIED, ALL METHODS USE A SEARCH SIZE OF 384

Method	AUC	$SR_{0.5}$	$SR_{0.75}$	Pre	Pre_N
Trackers with only backbone-pretrained initialization					
$OSTrack_{256}$ [29]	52.3	64.8	36.3	67.5	65.2
OSTrack [29]	55.1	69.6	44.1	73.3	68.8
ODTrack [60]	52.7	67.3	38.8	71.3	66.2
ZoomTrack [61]	55.7	70.5	45.7	74.9	69.6
UNTrack [5]	59.7	75.8	46.2	79.2	74.8
PEFT-based trackers with RGB tracking-pretrained initialization					
SDSTrack [22]	63.9	78.4	63.2	80.9	78.1
UnTrack [21]	64.5	78.8	65.1	80.6	78.9
BAT [62]	65.3	79.4	65.4	81.4	79.3
ViPT [20]	66.4	80.1	64.6	82.7	80.7
SSTtrack [10]	64.6	78.2	63.6	80.1	78.6
HDSP [18]	65.5	79.7	61.4	82.6	80.4
DaSSP [25]	66.2	80.0	64.4	82.9	80.3
SSTP (ours)	**69.2**	**83.4**	**66.7**	**85.6**	**83.4**

where $\mathcal{L}_{cls}$ is the focal loss for classification, while $\mathcal{L}_{giou}$ and $\mathcal{L}_1$ are employed for bounding box regression. Moreover, $\lambda_{giou} = 2$ and $\lambda_{\mathcal{L}_1} = 5$ serve as the regularization parameters.

IV. EXPERIMENTS

A. Implementation Details

1) Training: We conduct training and evaluation separately on each dataset, including MUST, HOT2020, and HOT2024. The resolution of the template image and search region is 192 × 192 and 384 × 384 pixels. We adopt the AdamW optimizer with an initial learning rate of 10^{-4}. We train for 15 epochs with a batch size of 32 and reduce the learning rate by a factor of 10 after six epochs. Considering the scale of different datasets, each epoch samples 60k image pairs for MUST and HOT2024, and 15k image pairs for HOT2020. We employ OSTrack as our pretrained base model, keeping its parameters frozen during training while initializing all trainable parameters using the Xavier uniform initialization scheme. All experiments are conducted on two NVIDIA RTX3090 GPUs.

2) Inference: We propagate long-term memory prompt tokens across consecutive frames, enabling the network to capture and maintain temporal dependencies over extended sequences.

B. Results on the MUST Dataset

1) Main Results: MUST is the first large-scale MS UAV single object tracking dataset. It consists of 160 sequences for training and 90 sequences for testing. Each frame has a spatial resolution of 1200×900 pixels with eight spectral bands ranging from 390 to 950 nm.

Table I reports the comparison results on the MUST dataset. The compared methods are divided into two groups according to their initialization and tuning paradigms. The first group is mainly provided as reference comparisons to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:20:28 UTC from IEEE Xplore. Restrictions apply.

TABLE II

COMPARISON OF MODEL PARAMS, FLOPS, AND SPEED. THE SPEED OF ALL METHODS IS TESTED ON A SINGLE NVIDIA GEFORCE RTX 3090

Model	FPS	FLOPs	Params
HIPTrack [19]	45.8	66.9G	120.4M
ARTrack [63]	15.1	172.1G	181.0M
SDSTrack [22]	13.6	244.3G	107.8M
UNTrack [5]	38.0	106.9G	112.6M
SSTP(ours)	**48.6**	**53.4G**	**104.9M**

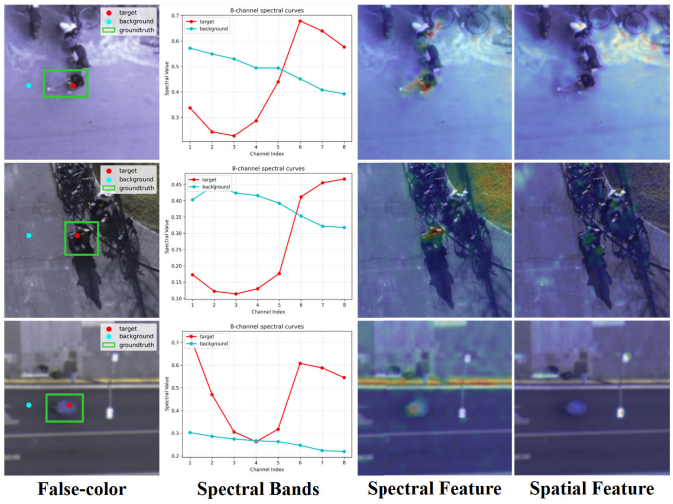


Fig. 5. Visualization of spatial and spectral feature maps shows that spectral features tend to focus on regions with high spectral distinctiveness.

show that inheriting the strong generalization ability of RGB trackers can generally improve the overall performance of MS tracking. The results are adopted from the original MUST paper, except for ZoomTrack, which we reproduced under the input resolution used in this work using the initialization and training strategy described in the original MUST paper. Therefore, our method is primarily compared with MS trackers following the PEFT-based paradigm. All methods in the second group are reproduced in our experimental setting by initializing from RGB trackers pretrained on COCO [56], LaSOT [57], GOT-10k [58], and TrackingNet [59], keeping the pretrained RGB tracker frozen, and optimizing only the newly introduced parameters.

As shown in Table I, SSTP achieves 69.2% AUC and 85.6% precision, outperforming strong PEFT-based trackers such as ViPT and DaSSP under the same search size. These results demonstrate the effectiveness of the proposed SSTP design, which fully exploits spectral cues and temporal dependencies for more robust MS tracking under limited training data.

2) Model Efficiency Analysis: We compare the Params, FLOPs, and speed with state-of-the-art trackers in Table II. By employing efficient dual-granularity spatial–spectral feature extraction and a prompt-based mechanism for temporal modeling, SSTP achieves robust tracking performance with relatively low computational costs while exceeding real-time speed requirements.

3) Visualization of Spectral Feature: To intuitively demonstrate how our model leverages spectral information to assist tracking, we visualize the spectral and spatial features within the DGS^{2}F module, as shown in Fig. 5. It can be observed that the refined spectral features tend to focus on regions

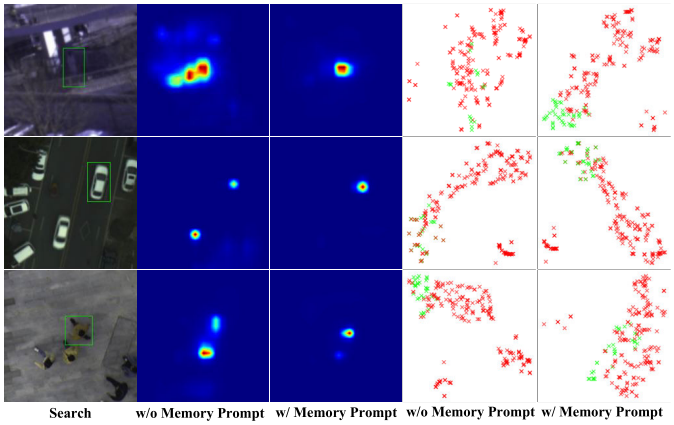


Fig. 6. Visualization of response and t-SNE maps. Temporal cues enhance the discriminability between target and background in challenging scenarios.

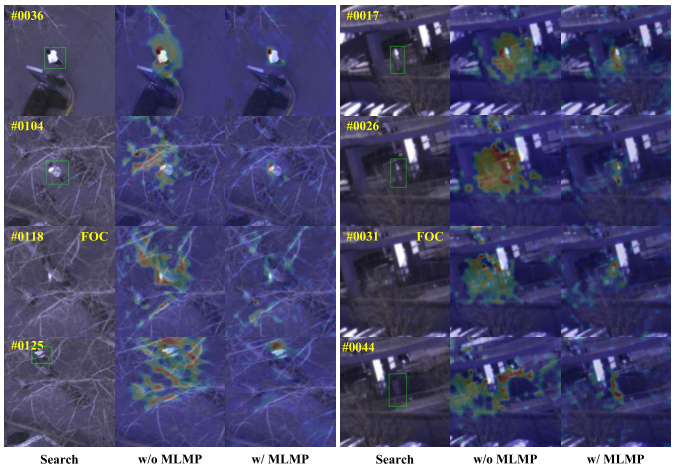


Fig. 7. Comparative visualization of attention maps with and without MLMP.

with salient spectral characteristics in the image, thereby compensating for the limitations of spatial features.

4) Visualization of Temporal Prompting: To provide an intuitive analysis of the proposed temporal prompting mechanism, we visualize the response maps, t-SNE feature distributions, and attention heatmaps under different temporal settings. As shown in Fig. 6, introducing temporal information helps the tracker better localize the target when target features are degraded by occlusion or disturbed by similar objects.

Fig. 7 further compares the attention heatmaps with and without MLMP on sequences involving long-term partial occlusion and full occlusion. Without MLMP, the attention easily drifts to background regions or distractors after occlusion. On the contrary, MLMP helps maintain attention on target-relevant regions and improves target recovery after reappearance. These visualizations demonstrate that the proposed temporal prompting mechanism effectively enhances robustness under challenging tracking scenarios.

5) Attribute-Based Comparison: Furthermore, we compare the performance of state-of-the-art PEFT-based MS trackers on each challenge attribute, as illustrated in Fig. 8. By extracting spatial and spectral information at dual granularities, SSTP fully exploits spectral cues and demonstrates superior performance over state-of-the-art MS trackers in scenarios where target appearance is severely constrained, such as IV, LR, and SC. Notably, by employing hierarchical temporal modeling,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:20:28 UTC from IEEE Xplore. Restrictions apply.

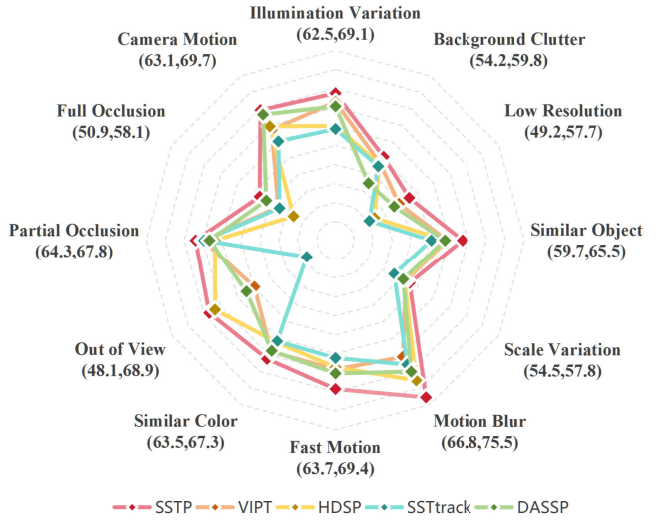


Fig. 8. Comparison of SSTP with other state-of-the-art PEFT-based MS trackers in terms of AUC across each challenge attribute on the MUST dataset, with the lowest and highest performance values marked.



Fig. 9. Visualization of tracking results on five video sequences (i.e., CM-Person-008, FOC-Person-013, LR-Person-001, LR-Person-005, and MB-Car-006). SSTP performs favorably against these PEFT-based MS trackers.

the comprehensive exploitation of temporal information enables the tracker to better discriminate between similar targets and handle extreme scenarios such as target occlusion. Under challenging scenarios such as SO, PO, and FO, SSTP consistently surpasses other MS trackers that also incorporate temporal information, including HDSP and SSTtrack.

6) Visualization of Tracking Results: We compare the tracking results of SSTP against various PEFT-based MS trackers, as shown in Fig. 9. In LR-Person-005 and MB-Car-006, the target is persistently surrounded by visually similar distractors and other challenging factors. Due to insufficient temporal modeling, other trackers tend to lose the target earlier, whereas SSTP maintains more accurate bounding boxes. In sequences with motion blur, low resolution, and occlusion, such as CM-Person-008 and FOC-Person-013, the effective use of spectral information further improves the separability between the target and the background. These visual results demonstrate that SSTP can jointly exploit spectral and temporal cues, resulting in more robust tracking over long temporal spans.

TABLE III

COMPARISON OF RESULTS FOR SSTP AND OTHER STATE-OF-THE-ART TRACKERS ON THE HOT2020 DATASET. FOR A FAIR COMPARISON, ALL TRACKERS PERFORM INFERENCE ON SINGLE NVIDIA RTX 3090 GPU

Method	AUC	Pre	FPS
MHT [33]	58.7	88.0	-
BAE-Net [12]	60.6	87.8	-
SSTNet [64]	62.1	90.1	-
SiamHYPER [65]	67.8	94.5	27.7
TMTNet [66]	69.9	92.8	12.6
SSTFT [16]	68.2	88.4	-
SEENet [67]	65.3	93.0	13.1
HANet [68]	68.8	94.8	21.2
UNTrack [5]	70.4	93.7	37.0
SSTP(ours)	**71.5**	**93.8**	**48.6**

C. Results on the HOT2020 Dataset

The HOT2020 [33] is the first public MS dataset for general tracking. It contains 40 sequences for training and 35 sequences for testing. Each frame contains 16 spectral bands ranging from 460 to 600 nm. We conducted training and evaluation on the HOT2020 dataset and compared the runtime performance with various MS trackers. As shown in Table III, SSTP achieves the fastest inference speed at 48 frames/s while also reaching state-of-the-art tracking accuracy. The leading performance in both accuracy and efficiency can be attributed to the full utilization of spectral and temporal cues, coupled with a concise single-stream architecture enabled by prompt learning.

D. Results on the HOT2024 Dataset

The HOT2024 benchmark is used to evaluate the proposed method. It was collected using three XIMEA snapshot cameras, including VIS, NIR, and RedNIR, which cover 16, 25, and 15 spectral bands after calibration, respectively. The training set contains 217 sequences, including 111 VIS, 70 NIR, and 36 RedNIR sequences. The validation set contains 117 sequences, including 67 VIS, 30 NIR, and 20 RedNIR sequences, denoted as VIS24, NIR24, and RedNIR24, respectively. As shown in Table IV, SSTP achieves leading performance compared with recent PEFT-based MS trackers, including PHTrack, SSTtrack, and SpectralTrack, demonstrating its effectiveness in adapting to MS data with different spectral ranges and numbers of bands after training on the HOT2024 dataset.

E. Ablation Study

To validate the effectiveness of the main components, we conduct rigorous ablation studies on the MUST dataset.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:20:28 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
COMPARISON OF RESULTS FOR SSTP AND OTHER STATE-OF-THE-ART TRACKERS ON THE HOT2024 DATASET

Method	NIR24		RedNIR24		VIS24	
	AUC	Pre	AUC	Pre	AUC	Pre
MHT [33]	40.5	73.1	23.8	38.3	38.4	56.4
BAE-Net [12]	46.3	76.7	30.7	42.9	32.9	49.0
SiamBAG [13]	53.6	78.3	22.6	35.8	39.7	55.9
SPIRIT [11]	65.6	82.4	37.9	54.5	32.3	42.1
SENSE [69]	56.2	76.6	36.7	49.6	30.1	41.0
Trans-DAT [15]	58.7	75.4	42.8	57.3	40.1	53.5
PHTrack [70]	52.8	73.2	26.4	39.5	30.6	42.3
SSTtrack [10]	71.1	88.9	51.1	68.3	38.6	49.0
SpectralTrack [47]	76.3	93.8	53.1	69.2	55.1	71.1
SSTP(ours)	**77.7**	**94.2**	**54.7**	**68.7**	**55.5**	**71.9**

TABLE V
INFLUENCE OF TRAINING STRATEGIES ON MUST

Method	Model Params	Tuned Params	AUC	Pre
Frozen	94.58M	1.57M	63.5	80.6
FFT	94.58M	94.58M	64.4	80.3
Adapter Tuning	101.70M	8.69M	65.5	81.4

TABLE VI
INFLUENCE OF ADAPTERS ON MUST. THE ABBREVIATIONS A AND P INDICATE THAT THE ADAPTER IS INSERTED AFTER THE CORRESPONDING OPERATION AND IN A PARALLEL POSITION, RESPECTIVELY

Position	$W_{Q/K/V}$	W_O	MSA	MLP	AUC	Pre
Extra-MSA	-	-	a	p	67.4	83.5
Intra-MSA	p	p	-	-	68.4	84.5
Intra-MSA	a	p	-	-	69.0	85.6
Intra-MSA	**a**	**a**	**-**	**-**	**69.2**	**85.6**

1) Effectiveness of the Adaptation: To evaluate the efficiency of adapter tuning, we employed two training strategies, freezing and FFT, on the foundation model. The comparative results are presented in Table V. In particular, under the frozen training strategy, standard patch embedding is separately applied to false-color and MS images, providing a reference for comparison. The extracted features are fused through elementwise summation before being fed into the backbone network, with only the MS patch embedding parameters being trainable. In the FFT experiment, all frozen parameters are unfrozen for training. On the contrary, adapter tuning introduces adapters based on the baseline model and only trains these adapters. According to the experimental results, the adapter tuning strategy outperforms both frozen training and FFT methods, demonstrating its effectiveness in transferring pretrained RGB knowledge to MS data. In addition, compared with FFT, it requires training far fewer parameters.

In addition, we investigate the impact of inserting adapters at different positions. For fairness, the total number of adapter parameters is kept consistent across all configurations. As shown in Table VI, inserting adapters inside the multihead self-attention (MSA) yields better performance than placing them externally. As a result, we adopt a design where adapters are distributed after the linear layers within the attention module for fine-grained adjustment.

TABLE VII
COMPONENT ANALYSIS ON MUST DATASET

Model	DGS^{2}F	SMP	MLMP	AUC	Pre
①	-	-	-	64.3	80.4
②	✓	-	-	66.3	82.0
③	✓	✓	-	67.9	84.1
④	✓	-	✓	68.0	84.3
SSTP	✓	✓	✓	**69.2**	**85.6**

TABLE VIII
ABLATION STUDY OF DIFFERENT SPECTRAL BRANCH DESIGNS AND FUSION STRATEGIES

Spa.	Spe.	Agg.	Fusion	AUC / Pre
✓	×	×	×	64.4 / 80.7
✓	16×16	×	Sum	65.5 / 81.4
✓	8×8	MaxPool	Sum	65.3 / 81.3
✓	4×4	MaxPool	Sum	65.8 / 81.7
✓	2×2	MaxPool	Sum	65.1 / 81.1
✓	4×4	Reorg.	Sum	66.0 / 81.8
✓	4×4	**Reorg.**	**Align**	**66.3 / 82.0**

Spa.: spatial branch; Spe.: spectral branch; Agg.: aggregation; Reorg.: spatial reorganization.

2) Component Analysis: Our model employs DGS^{2}F and an adapter-based transformer to effectively exploit spectral information, while progressively incorporating temporal cues via short-term memory prompting (SMP) and Mamba-based long-term memory prompting (MLMP). As shown in Table VII, we compare five different models. ① extracts spatial features only from false-color images and adopts the adapter tuning training strategy. Building upon ①, ② incorporates DGS^{2}F for spectral feature extraction and fusion. The performance improvement over ① demonstrates its enhanced capability to extract and utilize spectral information more effectively.

To investigate the value of temporal information incorporation, ③ introduces SMP. The performance improvement over ② demonstrates that the spectral–spatiotemporal feature extraction backbone generates more robust representations. ④ incorporates the MLMP, which aggregates critical information through tokens and propagates them across frames to establish long-range motion modeling. The performance gain over ② substantiates the value of long-term temporal information utilization. Compared to ③ and ④, SSTP integrates both spectral–spatiotemporal feature extraction with local memory and long-range motion modeling, achieving optimal performance through comprehensive utilization of spectral-spatiotemporal information in MS sequences.

3) Effectiveness of DGS^{2}F: To evaluate the key designs in DGS^{2}F, we conduct ablation experiments on different spectral branch configurations and spatial–spectral feature fusion strategies, as shown in Table VIII. The performance improvement achieved by introducing the original MS branch demonstrates the benefit of exploiting spectral cues beyond the false-color representation. Among different spectral patch sizes, the 4×4 configuration achieves the best performance,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:20:28 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
ABLATION STUDY OF TEMPORAL PROMPTING STRATEGIES FOR THE BACKBONE NETWORK

Hist.	Fusion	Residual	Params	AUC / Pre
×	×	×	0M	66.3 / 82.0
✓	Add	×	0.59M	66.7 / 82.8
✓	Concat	×	1.18M	66.2 / 82.4
✓	CompAdd	×	0.01M	66.8 / 82.8
✓	**CompAdd**	✓	**0.01M**	**67.9 / 84.1**

Hist.: historical token; CompAdd: compressed additive fusion in SMP.

TABLE X
INFLUENCE OF DIFFERENT SHORT-TERM MEMORY SPAN

#SMP Blocks	Span	Injection Layer	AUC	Pre
2	3	3,9	67.6	83.7
3	**4**	**3,6,9**	**69.2**	**85.6**
4	5	2,5,8,10	69.1	85.6
5	6	1,3,5,7,9	68.6	84.6

suggesting a favorable balance between fine-grained spectral modeling and local spatial consistency. The proposed spatial reorganization outperforms max pooling, and the alignment operation further enhances spatial–spectral fusion. These results validate the effectiveness of the key designs in DGS^{2}F.

4) Effectiveness of SMP: We further evaluate SMP from two aspects: the prompting structure and the number of inserted blocks. As shown in Table IX, we compare different structures, including direct addition, concatenation, compressed additive fusion, and the nonresidual variant. Compressed additive fusion achieves better performance with only 0.01M additional parameters per module, showing its ability to filter and aggregate valuable historical cues efficiently. The residual connection further improves the results, indicating that it is more suitable for the prompting paradigm by preserving the current prompt representation while incorporating complementary historical information.

If more SMP blocks are inserted into the network, the memory span of the backbone features increases accordingly. As shown in Table X, a smaller number of SMP blocks may lead to insufficient modeling capacity, while an excessive number of SMP blocks could introduce more early-stage targets during the feature extraction phase, potentially compromising the representation quality of the current target. Moreover, increasing the number of SMP blocks inevitably adds to the overall model parameters. Considering the tradeoff between performance and efficiency, we ultimately select three SMP blocks in our final design.

5) Effectiveness of MLMP: Table XI analyzes the influence of different long-term motion modeling strategies. Compared with long short-term memory (LSTM) and structured state space (S4) sequence models, Mamba consistently achieves superior AUC and precision scores, demonstrating stronger long-range dependence modeling capability. Compared with MSA, Mamba not only yields better tracking accuracy but also benefits from linear computational complexity, making it more scalable for long-term sequence modeling.

Furthermore, the sliding window length of the MLMP determines the temporal range over which the motion sequence is modeled, denoted as n. We evaluate the effect of different sliding window lengths by varying n while keeping the total batch size consistent across experiments to ensure fairness. These results demonstrate that a small window limits the model's ability to capture long-term dependencies, whereas a large window may overly emphasize outdated target information, thereby impairing the perception of the current target state. Therefore, the optimal window length is likely to be around 8. Considering that longer window lengths introduce additional computational overhead, we adopt a window length of 8 in the final model to balance effectiveness and efficiency.

TABLE XI
INFLUENCE OF LONG-TERM MOTION MODELING VARIANTS AND WINDOW LENGTH ON THE MUST

Variant	n	AUC	$SR_{0.5}$	$SR_{0.75}$	Pre	Pre_N
MSA	8	68.4	82.4	66.0	84.7	82.5
LSTM	8	68.1	82.3	65.6	84.6	81.7
S4	8	67.5	81.1	65.8	83.3	80.9
Mamba	4	68.5	82.6	66.0	84.8	82.5
Mamba	**8**	**69.2**	**83.4**	**66.7**	**85.6**	**83.4**
Mamba	16	68.1	82.1	65.6	84.0	82.2

TABLE XII
ROBUSTNESS ANALYSIS UNDER INACCURATE HISTORICAL PROMPTS ON THE MUST DATASET. WE REPLACE THE TEMPORAL TOKEN WITH GAUSSIAN NOISE EVERY K FRAMES DURING INFERENCE

Interval K	AUC	Pre	ΔAUC	ΔPre
Clean	69.2	85.6	-	-
24	68.7	84.9	-0.5	-0.7
16	68.2	84.3	-1.0	-1.3
8	67.9	84.1	-1.3	-1.5
6	67.7	83.9	-1.5	-1.7
4	67.4	83.4	-1.8	-2.2
2	67.2	83.2	-2.0	-2.4

6) Robustness to Inaccurate Historical Prompts: We first conduct a quantitative analysis to evaluate the robustness of SSTP to inaccurate historical prompts. To simulate the interference caused by low-quality features, we replace the temporal token input to MLMP with a Gaussian noise token every K frames during inference, where each element of the noise token is sampled from $\mathcal{N}(0, 1)$. As shown in Table XII, SSTP degrades gradually as the perturbation frequency increases. The performance drops only slightly when noise is injected every 24 or 16 frames/s, and even under the strongest setting ($K = 2$), SSTP still achieves 67.2% AUC and 83.2% precision. This indicates that inaccurate temporal prompts can affect long-term memory propagation, but SSTP is not overly sensitive to occasional corrupted prompts and degrades progressively rather than collapsing abruptly under continuous perturbations.

It is worth noting that the Gaussian perturbation is intended to provide a controlled and quantitative simulation of corrupted historical prompts, although it does not fully reproduce temporally correlated tracking drift or incorrect target updates that may occur in practical tracking scenarios. To intuitively demonstrate the effectiveness of the temporal design and its robustness under challenging conditions, Fig. 10 visualizes the IoU overlap curves and tracking results of SSTP and its variant without temporal information. The variant without

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:20:28 UTC from IEEE Xplore. Restrictions apply.

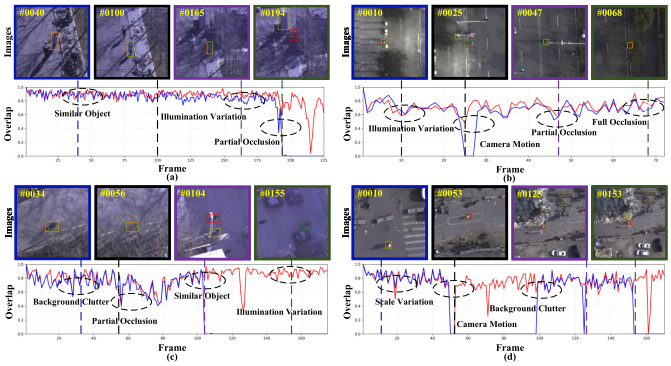


Fig. 10. Overlap curves and tracking results of SSTP and its variant without temporal information. Red curves and boxes denote SSTP, blue curves and yellow boxes denote the variant, and green boxes denote the ground truth. (a) MUST (BC-Person-016). (b) MUST (IV-Person-002). (c) MUST (IV-Person-007). (d) MUST (SV-Person-007).

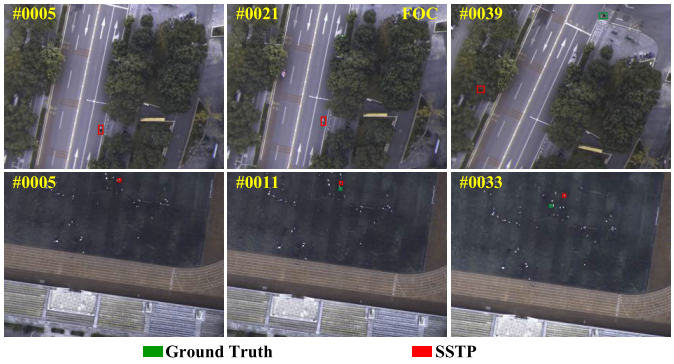


Fig. 11. Failure cases of SSTP (i.e., FOC-Motorcycle-001 and LR-Person-009).

temporal information is more likely to drift when the target is occluded or when abrupt camera motion causes severe blur. On the contrary, SSTP maintains higher overlap scores and more accurate bounding boxes under these challenging conditions. These visualizations demonstrate that the proposed temporal design improves tracking stability and robustness over long temporal spans.

F. Limitation and Discussion

Although SSTP is robust in most challenging scenarios, it can still fail under extreme conditions. As shown in Fig. 11, in FOC-Motorcycle-001, the target is fully occluded while moving rapidly, and a similar vehicle appears near its previous location. Due to the local-search paradigm, the tracker may remain around the previous target region and fail to recover the target after it moves far outside the search area. This issue could be mitigated by abnormal-state detection and full-frame redetection. In LR-Person-009, the target is extremely small and surrounded by similar pedestrians, making its representation ambiguous and leading to drift. Future work may address this by introducing lightweight motion prediction or multicandidate trajectory selection, which can better constrain target displacement and reduce drifting to nearby distractors.

V. Conclusion

In this article, we propose the SSTP, a spectral–spatiotemporal prompting framework for MS UAV tracking. The core idea of SSTP is to fully exploit the rich spatial–spectral information and temporal cues present in MS tracking videos. SSTP is designed with dual-granularity spatial–spectral feature fusion and a hierarchical temporal modeling mechanism, enabling effective utilization of both spectral and temporal information. Through separate training and evaluation on multiple MS datasets, SSTP demonstrates its applicability to different spectral configurations while achieving superior tracking accuracy and efficiency. We believe that SSTP will facilitate future progress in MS UAV tracking.

References

[1] J. Li, X. Liang, S. Shen, T. Xu, J. Feng, and S. Yan, "Scale-aware fast R-CNN for pedestrian detection," *IEEE Trans. Multimedia*, vol. 20, no. 4, pp. 985–996, Apr. 2018.

[2] J. Li, X. Liang, Y. Wei, T. Xu, J. Feng, and S. Yan, "Perceptual generative adversarial networks for small object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 1222–1230.

[3] Y. Xiao, T. Xu, X. Yu, Y. Fang, and J. Li, "A lightweight fusion strategy with enhanced interlayer feature correlation for small object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4708011.

[4] Y. Xiao, T. Xu, Y. Xin, and J. Li, "FBRT-YOLO: Faster and better for real-time aerial image detection," in *Proc. AAAI Conf. Artif. Intell.*, 2025, vol. 39, no. 8, pp. 8673–8681.

[5] H. Qin, T. Xu, T. Li, Z. Chen, T. Feng, and J. Li, "MUST: The first dataset and unified framework for multispectral UAV single object tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 16882–16891.

[6] Y. Tang, Y. Liu, L. Ji, and H. Huang, "Robust hyperspectral object tracking by exploiting background-aware spectral information with band selection network," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[7] S. Wang, K. Qian, and P. Chen, "BS-SiamRPN: Hyperspectral video tracking based on band selection and the Siamese region proposal network," in *Proc. 12th Workshop Hyperspectral Imag. Signal Process.*, Evol. Remote Sens., Sep. 2022, pp. 1–8.

[8] M. A. Islam, J. Zhou, W. Xing, Y. Gao, and K. K. Paliwal, "UBSTrack: Unified band selection and multimodel ensemble for hyperspectral object tracking," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5515515.

[9] Y. Wang, S. Mei, M. Ma, Y. Liu, and Y. Su, "HTACPE: A hybrid transformer with adaptive content and position embedding for sample learning efficiency of hyperspectral tracker," *IEEE Trans. Multimedia*, vol. 27, pp. 2384–2398, 2025.

[10] Y. Chen et al., "SSTtrack: A unified hyperspectral video tracking framework via modeling spectral–spatial–temporal conditions," *Inf. Fusion*, vol. 114, Feb. 2025, Art. no. 102658.

[11] Y. Chen, Q. Yuan, Y. Tang, Y. Xiao, J. He, and L. Zhang, "SPIRIT: Spectral awareness interaction network with dynamic template for hyperspectral object tracking," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5503116.

[12] Z. Li, F. Xiong, J. Zhou, J. Wang, J. Lu, and Y. Qian, "BAE-Net: A band attention aware ensemble network for hyperspectral object tracking," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2020, pp. 2106–2110.

[13] W. Li, Z. Hou, J. Zhou, and R. Tao, "SiamBAG: Band attention grouping-based Siamese object tracking network for hyperspectral videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514712.

[14] G. He, L. Gao, L. Chen, Y. Jiang, W. Xie, and Y. Li, "Hyperspectral object tracking with spectral information prompt," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 12, pp. 12636–12651, Dec. 2025.

[15] Y. Wu, L. Jiao, X. Liu, F. Liu, S. Yang, and L. Li, "Domain adaptation-aware transformer for hyperspectral object tracking," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 9, pp. 8041–8052, Sep. 2024.

[16] Y. Wang, Y. Liu, M. Ma, and S. Mei, "A spectral–spatial transformer fusion method for hyperspectral video tracking," *Remote Sens.*, vol. 15, no. 7, p. 1735, Mar. 2023.

[17] D. Zhao et al., "OCSCNet-tracker: Hyperspectral video tracker based on octave convolution and spatial–spectral capsule network," *Remote Sens.*, vol. 17, no. 4, p. 693, Feb. 2025.

[18] R. Yao, L. Zhang, Y. Zhou, H. Zhu, J. Zhao, and Z. Shao, "Hyperspectral object tracking with dual-stream prompt," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5500612.

[19] W. Cai, Q. Liu, and Y. Wang, "HIPTrack: Visual tracking with historical prompts," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 19258–19267.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:20:28 UTC from IEEE Xplore. Restrictions apply.

[20] J. Zhu, S. Lai, X. Chen, D. Wang, and H. Lu, "Visual prompt multi-modal tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 9516–9526.

[21] Z. Wu et al., "Single-model and any-modality for video object tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 19156–19166.

[22] X. Hou et al., "SDSTrack: Self-distillation symmetric adapter learning for multi-modal visual object tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 26551–26561.

[23] L. Hong et al., "OneTracker: Unifying visual object tracking with foundation models and efficient tuning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 19079–19091.

[24] H. Xing, W. Wei, L. Zhang, and C. Ding, "A visual prompt learning network for hyperspectral object tracking," *Pattern Recognit. Lett.*, vol. 196, pp. 59–65, Oct. 2025.

[25] Z. Li et al., "Multi-domain universal representation learning for hyperspectral object tracking," *Pattern Recognit.*, vol. 162, Jun. 2025, Art. no. 111389.

[26] Z. Wang, J. Luo, J. Liu, and X. Liao, "A one-step transformer-based adaptive tracking network for hyperspectral videos," *Signal, Image Video Process.*, vol. 19, no. 11, p. 936, Nov. 2025.

[27] L. Bertinetto, J. Valmadre, J. F. Henriques, A. Vedaldi, and P. H. S. Torr, "Fully-convolutional Siamese networks for object tracking," in *Proc. ECCV Workshops*. Springer, Oct. 2016, pp. 850–865.

[28] B. Li, J. Yan, W. Wu, Z. Zhu, and X. Hu, "High performance visual tracking with Siamese region proposal network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8971–8980.

[29] B. Ye, H. Chang, B. Ma, S. Shan, and X. Chen, "Joint feature learning and relation modeling for tracking: A one-stream framework," in *Proc. 17th Eur. Conf. Comput. Vis. (ECCV)*. Springer, Oct. 2022, pp. 341–357.

[30] D. Yuan, X. Chang, Z. Li, and Z. He, "Learning adaptive spatial–temporal context-aware correlation filters for UAV tracking," *ACM Trans. Multimedia Comput., Commun., Appl.*, vol. 18, no. 3, pp. 1–18, Aug. 2022.

[31] Y. Liang et al., "Spatial context-aware UAV target tracking for consumer electronics," *IEEE Trans. Consum. Electron.*, vol. 72, no. 1, pp. 546–557, Feb. 2026.

[32] S. Li, X. Yang, X. Wang, D. Zeng, H. Ye, and Q. Zhao, "Learning target-aware vision transformers for real-time UAV tracking," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4705718.

[33] F. Xiong, J. Zhou, and Y. Qian, "Material based object tracking in hyperspectral videos," *IEEE Trans. Image Process.*, vol. 29, pp. 3719–3733, 2020.

[34] K. Qian, J. Zhou, F. Xiong, H. Zhou, and J. Du, "Object tracking in hyperspectral videos with convolutional features and kernelized correlation filter," in *Proc. Int. Conf. Smart Multimedia*. Cham, Switzerland: Springer, 2018, pp. 308–319.

[35] B. Uzkent, A. Rangnekar, and M. J. Hoffman, "Tracking in aerial hyperspectral videos using deep kernelized correlation filters," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 449–461, Jan. 2019.

[36] J. Chen, K. Qian, W. Song, Y. Qiu, and S. Wang, "SiamTU: Hyperspectral video tracking against appearance changes using improved SiameGAT with adaptive template updating," *Infr. Phys. Technol.*, vol. 147, Jun. 2025, Art. no. 105811.

[37] J. Gao, T. Han, Y. Yuan, and Q. Wang, "Domain-adaptive crowd counting via high-Quality image translation and density reconstruction," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 8, pp. 4803–4815, Aug. 2023.

[38] J. Ye, C. Fu, G. Zheng, D. P. Paudel, and G. Chen, "Unsupervised domain adaptation for nighttime aerial tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 8896–8905.

[39] B. Lester, R. Al-Rfou, and N. Constant, "The power of scale for parameter-efficient prompt tuning," 2021, *arXiv:2104.08691*.

[40] N. Houlsby et al., "Parameter-efficient transfer learning for NLP," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 2790–2799.

[41] X. Lisa Li and P. Liang, "Prefix-Tuning: Optimizing continuous prompts for generation," 2021, *arXiv:2101.00190*.

[42] E. J. Hu et al., "LoRA: Low-rank adaptation of large language models," in *Proc. ICLR*, 2022, vol. 1, no. 2, p. 3.

[43] D. Zhang, B. Li, F. Wang, Z. Zhao, and J. Gao, "Boosting quantitive and spatial awareness for zero-shot object counting," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2026, pp. 20356–20366.

[44] D. Liao et al., "Fine-grained feature and template reconstruction for TIR object tracking," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 9, pp. 9276–9286, Sep. 2025.

[45] M. V. Koroteev, "BERT: A review of applications in natural language processing and understanding," 2021, *arXiv:2103.11943*.

[46] B. Kang et al., "UETrack: A unified and efficient framework for single object tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2026, pp. 20890–20901.

[47] Y. Chen et al., "Hyperspectral video tracking with spectral–spatial fusion and memory enhancement," *IEEE Trans. Image Process.*, vol. 34, pp. 3547–3562, 2025.

[48] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," 2021, *arXiv:2111.00396*.

[49] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," in *Proc. 1st Conf. Lang. Model. (COLM)*, 2024, pp. 1–32.

[50] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, "Vision mamba: Efficient visual representation learning with bidirectional state space model," 2024, *arXiv:2401.09417*.

[51] Y. Liu et al., "VMamba: Visual state space model," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 37, 2024, pp. 103031–103063.

[52] A. Hatamizadeh and J. Kautz, "MambaVision: A hybrid mamba-transformer vision backbone," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 25261–25270.

[53] B. Kang, X. Chen, S. Lai, Y. Liu, Y. Liu, and D. Wang, "Exploring enhanced contextual information for video-level object tracking," in *Proc. AAAI Conf. Artif. Intell.*, 2025, vol. 39, no. 4, pp. 4194–4202.

[54] X. Li, B. Zhong, Q. Liang, G. Li, Z. Mo, and S. Song, "MambaLCT: Boosting tracking via long-term context state space model," in *Proc. AAAI Conf. Artif. Intell.*, 2025, vol. 39, no. 5, pp. 4986–4994.

[55] C. Xiao, Q. Cao, Z. Luo, and L. Lan, "MambaTrack: A simple baseline for multiple object tracking with state space model," in *Proc. 32nd ACM Int. Conf. Multimedia*, Oct. 2024, pp. 4082–4091.

[56] T.-Y. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis. Cham*, Switzerland: Springer, 2014, pp. 740–755.

[57] H. Fan et al., "LaSOT: A high-Quality benchmark for large-scale single object tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5374–5383.

[58] L. Huang, X. Zhao, and K. Huang, "GOT-10k: A large high-diversity benchmark for generic object tracking in the wild," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 5, pp. 1562–1577, May 2021.

[59] M. Müller, A. Bibi, S. Giancola, S. Alsubaihi, and B. Ghanem, "TrackingNet: A large-scale dataset and benchmark for object tracking in the wild," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Jun. 2018, pp. 300–317.

[60] Y. Zheng, B. Zhong, Q. Liang, Z. Mo, S. Zhang, and X. Li, "ODTrack: Online dense temporal token learning for visual tracking," in *Proc. AAAI Conf. Artif. Intell.*, 2024, pp. 7588–7596.

[61] Y. Kou et al., "ZoomTrack: Target-aware non-uniform resizing for efficient visual tracking," in *Proc. Adv. Neural Inf. Process. Syst.* 36, 2023, pp. 50959–50977.

[62] B. Cao, J. Guo, P. Zhu, and Q. Hu, "Bi-directional adapter for multi-modal tracking," in *Proc. AAAI Conf. Artif. Intell.*, 2024, pp. 927–935.

[63] X. Wei, Y. Bai, Y. Zheng, D. Shi, and Y. Gong, "Autoregressive visual tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 9697–9706.

[64] Z. Li, X. Ye, F. Xiong, J. Lu, J. Zhou, and Y. Qian, "Spectral–spatial–temporal attention network for hyperspectral tracking," in *Proc. 11th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens.* (WHISPERS), Mar. 2021, pp. 1–5.

[65] Z. Liu, X. Wang, Y. Zhong, M. Shu, and C. Sun, "SiamHYPER: Learning a hyperspectral object tracker from an RGB-based tracker," *IEEE Trans. Image Process.*, vol. 31, pp. 7116–7129, 2022.

[66] C. Zhao, H. Liu, N. Su, C. Xu, Y. Yan, and S. Feng, "TMTNet: A transformer-based multimodality information transfer network for hyperspectral object tracking," *Remote Sens.*, vol. 15, no. 4, p. 1107, Feb. 2023.

[67] Z. Li, F. Xiong, J. Zhou, J. Lu, and Y. Qian, "Learning a deep ensemble network with band importance for hyperspectral object tracking," *IEEE Trans. Image Process.*, vol. 32, pp. 2901–2914, 2023.

[68] Z. Liu, Y. Zhong, G. Ma, X. Wang, and L. Zhang, "A deep temporal-spectral–spatial anchor-free Siamese tracking network for hyperspectral video object tracking," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5539216.

[69] Y. Chen, Q. Yuan, Y. Tang, Y. Xiao, J. He, and Z. Liu, "SENSE: Hyperspectral video object tracker via fusing material and motion cues," *Inf. Fusion*, vol. 109, Sep. 2024, Art. no. 102395.

[70] Y. Chen et al., "PHTrack: Prompting for hyperspectral video tracking," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5533918.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:20:28 UTC from IEEE Xplore. Restrictions apply.